



02451 Introduction to Machine Learning

Week 2: Summary statistics, similarity, and nearest neighbors

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10 February 2026

DTU Compute, Technical University of Denmark



Today

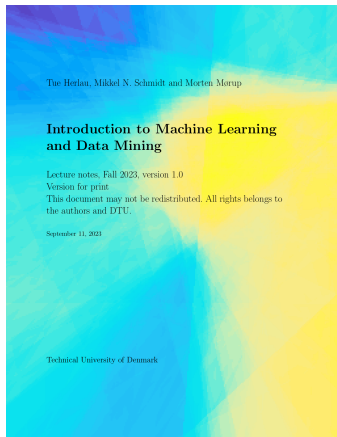
Feedback Groups of the day:

Pedro Miguel Apolinário Correia, Magnus Axelsson, Gideon Sylvester Lyhne Christensen, Jacob Levi Edsberg, Frederik Elm-Fich, Nikolaj Christian Fallenkamp, Emil Frydensberg, Morten Hedelykke Dietz Fuglsang, Wen Gao, Noam Emil Smitt Gil, Hannah Hammami Hauge, Setare Izadi, Carsten Uldahl Jansen, Linxin Jing, Magnus Berg Joensen, Frederik Jørgensen, Dines Peter Fink Kjeldgaard, Morten Kokholm Kristensen, Rasmus Helvig Larsen, Carlos Horácio Ferreira Sousa Lemos, Gabriel Nicholas Pryszlak Loza, Esben Mark-Hansen, Simon Allermann Møller, Elena Soulas Moreno, Jude Mumm, Rasmus Søgaard Nielsen, Leonor Nunes Fialho de Sousa Castelo, Emma Svane Olsen, Peter Adrian Petersen, Philip Rying, Balthasar Graff Stensballe, Fedir Vasyliiev, Daniel Yttesen

Reading/homework material:

Chapter 4, Chapter 12

P4.2, P4.3, P4.5, P12.1



Lecture Schedule

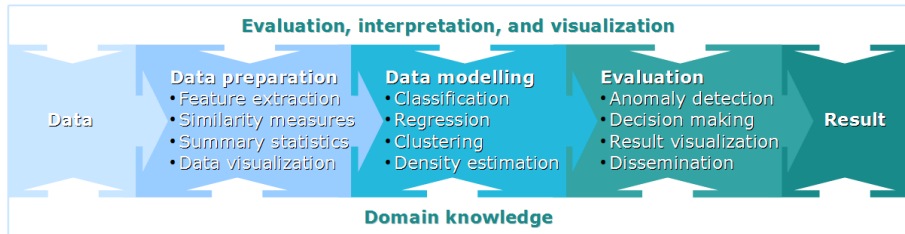
- 1 Introduction, data, and visualization
3 February: C1, C2, C7
- 2 **Summary statistics, similarity, and nearest neighbors**
10 February: C4, C12
- 3 Computational linear algebra and PCA
17 February: C3
- 4 Probability theory, Bayes and Naive Bayes
24 February: C5, C6, C13 (Assignment 1-3 due on 26 February)
- 5 Bayesian inference and linear models
3 March: C5, C6, C8
- 6 Decision trees, overfitting and cross-validation
10 March: C9, C10
- 7 Performance evaluation and AUC
17 March: C11, C16 (Assignment 4-6 due on 19 March)
- 8 Artificial Neural Networks and Optimization
24 March: C15
- 9 Bias/Variance and ensemble methods
7 April: C14, 17
- 10 K-means and hierarchical clustering
14 April: C18 (Assignment 7-9 due on 16 April)
- 11 Mixture models and density estimation
21 April: C19, C20
- 12 Representation learning
28 April: Material provided separately
- 13 Recap and discussion of the exam
5 May: C1-C20 (Assignment 10-12 due on 7 May)

Online help: Piazza

Videos of lectures: <https://panopto.dtu.dk>

Streaming of lectures: Zoom (link on DTU Learn)

Learning Objectives



Learning Objectives

- Understand and apply a range of basic statistics for summarizing a dataset
- Compute measures of similarity/dissimilarity (Lp distance, cosine distance, etc.)
- Explain how K-Nearest Neighbours can be used in classification and regression
- Evaluate classifiers in terms of the confusion matrix, error rate and accuracy

Plan for today:

- Lecture 2 (13:00 – ~15:00)
 - **Walk-through of assignment format and formalities (live)**
 - **Data revisited**
 - **Summary statistics**
 - **Break**
 - **Measures of similarity and dissimilarity**
 - **Nearest Neighbor methods for supervised learning**
- Exercises (15:00–17:00)

Practicalities and announcements

- Lecture rooms (overflow in A082; not A083)
- Mandatory attendance at lecture and exercises?
- Exam date (30/5 2026, Saturday):
`https://student.dtu.dk/en/exam/exam-dates#ordinary-summer-exam-and-re-exam-may--june-2026`
- 2 sheets (4 pages) of **handwritten** notes (not printed iPad/tablet notes)
- Please use Piazza or 02451@compute.dtu.dk to contact us

• HTML!

Data

- **Dataset types:** Tabular, relational, sequence
- **Data issues:** Missing, noisy, quality.
- **Attribute types:** Nominal, Ordinal, Interval, Ratio
- **Transformations** of attributes

- **Discrete**
 - Finite (or countably infinite) set of values
 - **Continuous**
 - Real number
-
- **Nominal** (Equal / Not equal)
 - Objects belong to a category
 - **Ordinal** (Greater than / Less than)
 - Objects can be ranked
 - **Interval** (Addition / Subtraction)
 - Distance between objects can be measured
 - **Ratio** (Multiplication / Division)
 - Zero means absence of what is measured

How do we represent, summarise, validate, and understand data before we can carry out machine learning and modelling?

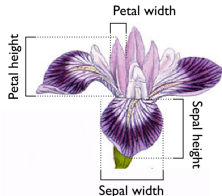
Data points and matrices

A data point is a set of **real** numbers represented as a vector

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_M \end{bmatrix} \in \mathbb{R}^M \quad \text{e.g. } \mathbf{x} = \begin{bmatrix} 5.1 \\ 3.5 \\ 1.4 \\ 0.2 \end{bmatrix}$$

A matrix is used to hold multiple data points (as rows)

$$\mathbf{X} = \begin{bmatrix} x_{1,1} & \dots & x_{1,M} \\ \vdots & & \vdots \\ x_{N,1} & \dots & x_{N,M} \end{bmatrix} \in \mathbb{R}^{N \times M}$$



Sepal Length	Sepal Width	Petal Length	Petal Width
5.1	3.5	1.4	0.2
4.9	3.0	1.4	0.2
4.7	3.2	1.3	0.2
4.6	3.1	1.5	0.2
5.8	2.7	5.1	1.9
$\vdots$	$\vdots$	$\vdots$	$\vdots$
5.7	2.8	4.1	1.3

$$\mathbf{X} = \begin{bmatrix} 5.1 & 3.5 & 1.4 & 0.2 \\ 4.9 & 3.0 & 1.4 & 0.2 \\ 4.7 & 3.2 & 1.3 & 0.2 \\ 4.6 & 3.1 & 1.5 & 0.2 \\ 5.8 & 2.7 & 5.1 & 1.9 \\ \vdots & \vdots & \vdots & \vdots \\ 5.7 & 2.8 & 4.1 & 1.3 \end{bmatrix}$$

Matrix basics

- Common matrix notation

$\mathbf{A}$, $\mathbf{A}$, A , $\overline{\overline{A}}$

$$\mathbf{A} = \begin{bmatrix} a_{1,1} & \cdots & a_{1,M} \\ \vdots & & \vdots \\ a_{N,1} & \cdots & a_{N,M} \end{bmatrix} \in \mathbb{R}^{N \times M}$$

- Common vector notation

$\mathbf{x}$, $\mathbf{x}$, x , $\overline{x}$, $\vec{x}$

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_M \end{bmatrix} \in \mathbb{R}^M$$

Matrix multiplication

$$\mathbf{A} \times \mathbf{B} = \mathbf{C}$$

$$L \times M \quad M \times N \quad L \times N$$

$$\begin{array}{c} 3 \times 4 \text{ matrix} \\ \left[\begin{array}{cccc} \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ \mathbf{1} & \mathbf{2} & \mathbf{3} & \mathbf{4} \end{array} \right] \end{array} \begin{array}{c} 4 \times 5 \text{ matrix} \\ \left[\begin{array}{ccccc} \cdot & \cdot & \cdot & \mathbf{a} & \cdot \\ \cdot & \cdot & \cdot & \mathbf{b} & \cdot \\ \cdot & \cdot & \cdot & \mathbf{c} & \cdot \\ \cdot & \cdot & \cdot & \mathbf{d} & \cdot \end{array} \right] \end{array} = \begin{array}{c} 3 \times 5 \text{ matrix} \\ \left[\begin{array}{ccccc} \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & x_{3,4} & \cdot \end{array} \right] \end{array}$$

$$x_{3,4} = \mathbf{1} \cdot \mathbf{a} + \mathbf{2} \cdot \mathbf{b} + \mathbf{3} \cdot \mathbf{c} + \mathbf{4} \cdot \mathbf{d}$$

Matrix transpose

- The transpose of a matrix

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 4 & 6 \\ 7 & 8 & 9 \end{bmatrix} \quad \mathbf{A}^\top = \begin{bmatrix} 1 & 3 & 7 \\ 2 & 4 & 8 \\ 3 & 6 & 9 \end{bmatrix}$$

- Transpose of a sum

$$(\mathbf{A} + \mathbf{B})^\top = \mathbf{A}^\top + \mathbf{B}^\top$$

- Transpose of a product

$$(\mathbf{AB})^\top = \mathbf{B}^\top \mathbf{A}^\top$$

$$(\mathbf{Ax})^\top \mathbf{y} = \mathbf{x}^\top \mathbf{A}^\top \mathbf{y} = \mathbf{x}^\top (\mathbf{A}^\top \mathbf{y})$$

Matrix inverse

- Ones on the diagonal and zeros everywhere else

$$\mathbf{I} = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix} \quad \mathbf{I}^\top = \mathbf{I}$$

- Multiplying by the identity does not change anything

$$\mathbf{I}\mathbf{A} = \mathbf{A}$$

$$\mathbf{I}_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$\mathbf{I}_2\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

- For a square matrix, the inverse satisfies

$$\mathbf{A}\mathbf{A}^{-1} = \mathbf{A}^{-1}\mathbf{A} = \mathbf{I}$$

Scaling, rotation and shear

Rotation:

$$R(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

$$x' = Rx$$

E.g. $R = \begin{bmatrix} 0.87 & -0.5 \\ 0.5 & 0.87 \end{bmatrix}$

Scaling and stretching:

$$S(\alpha_x, \alpha_y) = \begin{bmatrix} \alpha_x & 0 \\ 0 & \alpha_y \end{bmatrix}$$

$$x' = Sx$$

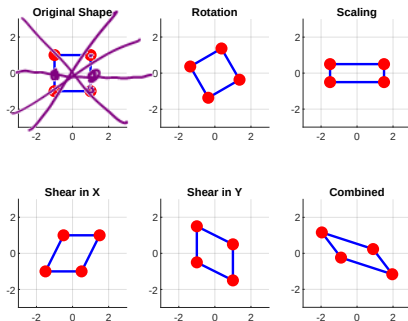
E.g. $S = \begin{bmatrix} 1.5 & 0 \\ 0 & 0.5 \end{bmatrix}$

Shear:

$$D_x(\alpha_x, \alpha_y, \beta) = \begin{bmatrix} \alpha_x & \beta \\ 0 & \alpha_y \end{bmatrix}$$

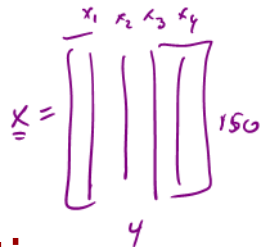
$$x' = D_x x$$

E.g. $D_x = \begin{bmatrix} 1 & 0.5 \\ 0 & 1 \end{bmatrix}$

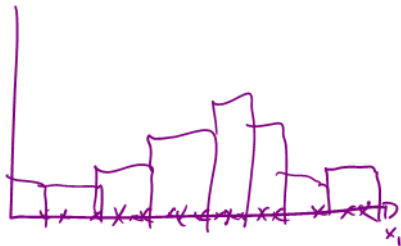


$$x' = Sx$$





Summary Statistics



Empirical statistics

Given two samples $x_1, x_2, \dots, x_N \in \mathbb{R}$ and $y_1, y_2, \dots, y_N \in \mathbb{R}$:

- Empirical mean

$$\hat{\mu} = \bar{x} = \frac{1}{N} \sum_{i=1}^N x_i$$

- Empirical variance

$$\hat{s} = \text{var}[x] = \frac{1}{N-1} \sum_{i=1}^N (x_i - \hat{\mu})^2$$

- Empirical covariance

$$\text{cov}[x, y] = \frac{1}{N-1} \sum_{i=1}^N (x_i - \hat{\mu}_x)(y_i - \hat{\mu}_y)$$

- Empirical standard deviation

$$\hat{\sigma} = \text{std}[x] = \sqrt{\hat{s}}$$



	Sepal Length	Sepal Width	Petal Length	Petal Width
$\mathbf{X} =$	5.1	3.5	1.4	0.2
	4.9	3.0	1.4	0.2
	4.7	3.2	1.3	0.2
	4.6	3.1	1.5	0.2
	5.8	2.7	5.1	1.9
	$\vdots$	$\vdots$	$\vdots$	$\vdots$
	5.7	2.8	4.1	1.3

Quantiles and percentiles

Given N observations of an attribute $x_1, x_2, \dots, x_N \in \mathbb{R}$.

Quantiles describe the *points* that divide the underlying distribution into intervals that are equally probable:

- The one 2-quantile (**median**) divides the distribution in two intervals.
- The three 4-quantiles (**quartiles**) divides the distribution in four intervals.
- The 99 100-quantiles (**percentiles**) divides the distribution in 100 intervals.

The **median** is the same as the 2nd quartile or the 50th percentile.

E.g., we can (approximately) find the **median** by

- Sort the observations in ascending order $x_{(1)} \leq x_{(2)} \leq \dots \leq x_{(N)}$

$$\text{median}[x] = \begin{cases} x_{(\frac{N+1}{2})} & \text{if } N \text{ is odd} \\ \frac{1}{2} \left(x_{(\frac{N}{2})} + x_{(\frac{N}{2}+1)} \right) & \text{if } N \text{ is even.} \end{cases}$$

Covariance

- Consider the case where we have more than two attributes and we want to estimate the (linear) relationship between all of them (pairwise).
- Empirical covariance

$$\text{cov}[\mathbf{x}^{(k)}, \mathbf{x}^{(m)}] = \frac{1}{N-1} \sum_{i=1}^N (x_i^{(k)} - \hat{\mu}_{x^{(k)}})(x_i^{(m)} - \hat{\mu}_{x^{(m)}})$$

$$\mathbf{X} = \begin{array}{c} \begin{array}{cccc} \text{Sepal Length} & \text{Sepal Width} & \text{Petal Length} & \text{Petal Width} \end{array} \\ \begin{bmatrix} 5.1 & 3.5 & 1.4 & 0.2 \\ 4.9 & 3.0 & 1.4 & 0.2 \\ 4.7 & 3.2 & 1.3 & 0.2 \\ 4.6 & 3.1 & 1.5 & 0.2 \\ 5.8 & 2.7 & 5.1 & 1.9 \\ \vdots & \vdots & \vdots & \vdots \\ 5.7 & 2.8 & 4.1 & 1.3 \end{bmatrix} \end{array}$$

- Empirical covariance matrix

$$\hat{\Sigma} = \begin{bmatrix} \text{cov}[x^{(1)}, x^{(1)}] & \dots & \text{cov}[x^{(1)}, x^{(M)}] \\ \vdots & & \vdots \\ \text{cov}[x^{(M)}, x^{(1)}] & \dots & \text{cov}[x^{(M)}, x^{(M)}] \end{bmatrix}$$

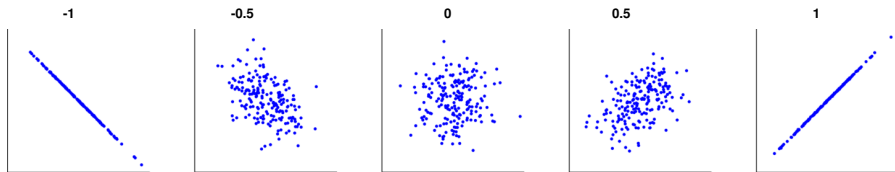
$$\hat{\Sigma} = \begin{bmatrix} 0.7 & 0.0 & 1.3 & 0.5 \\ 0.0 & 0.2 & -0.3 & -0.1 \\ 1.3 & -0.3 & 3.1 & 1.3 \\ 0.5 & -0.1 & 1.3 & 0.6 \end{bmatrix}$$

Correlation

- Measure of degree of linear relationship

$$\text{côr}[x, y] = \frac{\text{côv}[x, y]}{\hat{\sigma}_x \hat{\sigma}_y}$$

A correlation of 1 or -1 means that there is a perfect linear relation



1355

- TA Rooms

Dissimilarity and Similarity

Similarity / Dissimilarity measures

Similarity $s(x, y)$ Often between 0 and 1. Higher means more similar

Dissimilarity $d(x, y)$ Greater than 0. Lower means more similar.

Classification Classify a document as having the same topic y as the document it is **most similar/least dissimilar** to.

Outlier detection The observation most **dissimilar** to all other observations is an outlier



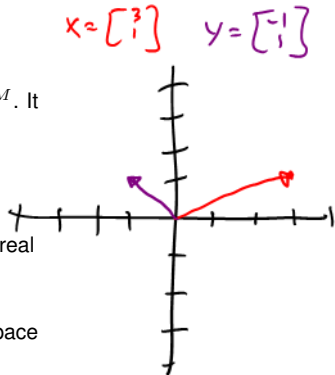
Vector spaces

Any vector of given dimension, M , lies in a vector space, called $\mathbb{R}^M$. It has the operations of:

- **scalar multiplication:** ax is defined for any scalar a .
For real vectors, $ax = [ax_1, ax_2, \dots, ax_M]$, i.e. element-wise scaling ($\mathbb{R} \times \mathbb{R}^M \rightarrow \mathbb{R}^M$)
- **vector addition:** $x + y$ vectors x, y of equal dimension. For real vectors, $x + y = [x_1 + y_1, x_2 + y_2, \dots, x_M + y_M]$ the element-wise sum ($\mathbb{R}^M \times \mathbb{R}^M \rightarrow \mathbb{R}^M$)
- **linear combinations** results in a vector in the same vector space (i.e. closed under linear combinations):
 $a_1x_1 + a_2x_2 + \dots + a_Nx_N$ for $a_k \in \mathbb{R}$

We will consider vector spaces which are equipped with two additional operations:

- **a norm** $\|x\|$ which allows the length of vectors to be measured.
($\mathbb{R}^M \rightarrow \mathbb{R}_{\geq 0}$)
- **an inner product** $\langle x, y \rangle$, $x^\top y$ or $x \cdot y$ allows the angles of two vectors to be compared. The inner product of two orthogonal vectors is 0. For real vectors $x^\top y = x_1y_1 + x_2y_2 + \dots + x_My_M$.
($\mathbb{R}^M \times \mathbb{R}^M \rightarrow \mathbb{R}$)



Norms: How big is that vector / matrix?

- The (Euclidian) norm of a vector measures it's length (magnitude):

$$\|\mathbf{x}\|_2 = \sqrt{\sum_{n=1}^N |x_n|^2} = \sqrt{\mathbf{x}^\top \mathbf{x}}$$

- The Frobenius norm of a matrix measures it's magnitude:

$$\|\mathbf{X}\|_F^2 = \sum_{i,j} x_{i,j}^2 = \text{trace}(\mathbf{X}\mathbf{X}^T) = \text{trace}(\mathbf{X}^T\mathbf{X})$$

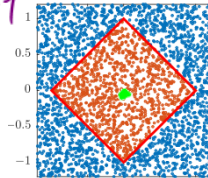
Where trace takes the sum of the diagonal elements, i.e. $\text{trace}(\mathbf{A}) = \sum_{i=1}^N a_{i,i}$

Dissimilarity measures

$$\mathbf{x} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \mathbf{y} = \begin{bmatrix} ? \\ ? \end{bmatrix} \quad p=1$$

- General Minkowsky distance (p -distance)

$$d_p(\mathbf{x}, \mathbf{y}) = \left(\sum_{j=1}^M |x_j - y_j|^p \right)^{\frac{1}{p}} = \|\mathbf{x} - \mathbf{y}\|_p$$



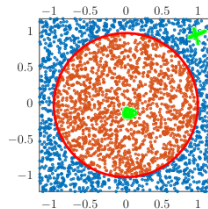
- One-norm ($p = 1$)

$$d_1(\mathbf{x}, \mathbf{y}) = \sum_{j=1}^M |x_j - y_j|$$

- Euclidean ($p = 2$)

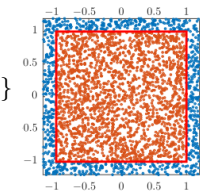
$$d_2(\mathbf{x}, \mathbf{y}) = \sqrt{\sum_{j=1}^M (x_j - y_j)^2}$$

$p=2$



- Max-norm distance ($p = \infty$)

$$d_\infty(\mathbf{x}, \mathbf{y}) = \max \{|x_1 - y_1|, |x_2 - y_2|, \dots, |x_M - y_M|\}$$



Usage: Regularization and alternative optimization targets.

For instance, $p = \infty$ is very affected by outliers, $p = 1$ much less so.

Quiz 1: Norms and distances

$$d_p(x, y) = \left(\sum_{j=1}^M |x_j - y_j|^p \right)^{1/p}$$

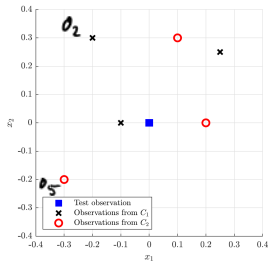


Figure 8: The synthetic dataset, where the colors indicate the classes such that the black observations $\{o_1, o_2, o_3\}$ belong to class C_1 , and the red observations $\{o_4, o_5, o_6\}$ belong to class C_2 .

	x_1	x_2
o_1	0.25	0.25
o_2	-0.2	0.3
o_3	-0.1	0.0
o_4	0.1	0.3
o_5	-0.3	-0.2
o_6	0.2	0.0

Consider a small synthetic dataset consisting of six observations and two attributes. The subset is presented in the figure on the left. We consider the p-distances d_1, d_2, d_∞ . A test point, o^* , is located at $[0, 0]^T$.

Determine which one of the following statements is correct.

A $d_1(o^*, o_3) > d_2(o^*, o_6)$

B $d_2(o_2, o_5) = 1$

C $\left\| o^* - \begin{bmatrix} -0.2 \\ 0.3 \end{bmatrix} \right\|_2^2 = 0.13$

D $d_p(o_4, o_k) > 0.1 \quad \forall k \in [1, 2, 3, 4, 5, 6]$
 $\forall p \in \{1, 2, \infty\}$

E Don't know

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} - \begin{bmatrix} -0.2 \\ 0.3 \end{bmatrix} = \begin{bmatrix} 0.2 \\ -0.3 \end{bmatrix}$$

$$= \frac{4}{100} + \frac{9}{100}$$

$$\left(\sqrt{(0.2)^2 + (-0.3)^2} \right)^2 = \left(\frac{2}{10} \right)^2 + \left(\frac{3}{10} \right)^2$$

$$\max\{0.1, 0.5\} = 0.5$$

Quiz 1: SOLUTION

- A $d_1(\mathbf{o}^*, \mathbf{o}_3) > d_2(\mathbf{o}^*, \mathbf{o}_6)$: Incorrect, for these two points we only need to consider distance along one axis making the problem much simpler. The value of p does not matter in this case, and we simply consider the absolute distance.
- B $d_\infty(\mathbf{o}_2, \mathbf{o}_5) = 1$: This is incorrect as the correct distance is 5 ($\max\{0.1, 0.5\}$)
- C $\left\| \mathbf{x}^* - \begin{bmatrix} -0.2 \\ 0.3 \end{bmatrix} \right\|_2^2 = \left(\sqrt{\sum_{j=1}^M (x_j - y_j)^2} \right)^2 = \sum_{j=1}^M (x_j - y_j)^2 = (-0.2)^2 + (0.3)^2 = 0.13$. Squaring the distance/norm will become important in future lectures.
- D $d_p(\mathbf{o}_4, \mathbf{o}_k) > 0.1 \ \forall k \in [1, 2, 3, 4, 5, 6] \ \forall p \in \{1, 2, \infty\}$: Incorrect, the distance from $\mathbf{o}_4$ to itself is 0 using any of the norms.
- E Don't know

Distance from (0,0) to obs.:

Obs	$p = 1$	$p = 2$	$p = \infty$
$\mathbf{o}_1$	0.5	0.3536	0.25
$\mathbf{o}_2$	0.5	0.3606	0.30
$\mathbf{o}_3$	0.1	0.1000	0.10
$\mathbf{o}_4$	0.4	0.3162	0.30
$\mathbf{o}_5$	0.5	0.3606	0.30
$\mathbf{o}_6$	0.2	0.2000	0.20

Similarity measures

Cosine similarity

$$x = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix} \quad y = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}$$

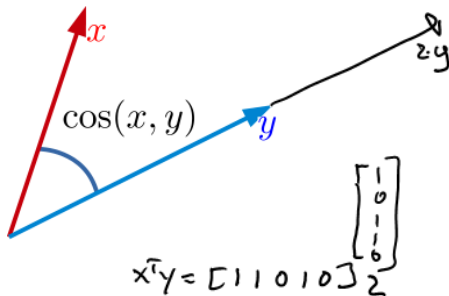
$$\cos(x, y) = \frac{x^T y}{\|x\|_2 \|y\|_2}$$

$$f_{11} = 2$$

$$f_{00} = 1$$

$$f_{01} = 1$$

$$f_{10} = 1$$



Similarity measures

— mostly for binary vectors

$$x = \begin{bmatrix} 0 \\ 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix} \quad y = \begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}$$

K : Total number of attributes
 f_{00} : Number of attributes where $x_k = y_k = 0$
 f_{11} : Number of attributes where $x_k = y_k = 1$

$$f_{01}$$
$$f_{10}$$

Simple Matching Coefficient (SMC)

$$\text{SMC}(x, y) = \frac{f_{00} + f_{11}}{K}$$

Jaccard Coefficient

$$J(x, y) = \frac{f_{11}}{K - f_{00}}$$

Extended Jaccard coefficient

$$\text{EJ}(x, y) = \frac{x^\top y}{\|x\|^2 + \|y\|^2 - x^\top y}$$

Quiz2: Similarity measures

Calculate the simple matching coefficient, Jaccard, cosine and extended jaccard similarity between customer 1 and customer 2 in the market basket data below.

$$x = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \\ 1 \end{bmatrix} \quad y = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \\ 1 \end{bmatrix}$$

ID	Bread	Soda	Milk	Beer	Diaper
1	1	1	1	0	0
2	0	1	1	0	1

$$f_{11} = 2$$

$$f_{00} = 1$$

$$f_{01} = 1$$

$$f_{10} = 1$$

K : Total number of attributes

f_{00} : Number of attributes where $x_k = y_k = 0$

f_{11} : Number of attributes where $x_k = y_k = 1$

$$SMC = \frac{1+2}{5} = \frac{3}{5}$$

$$J = \frac{2}{4} = \frac{1}{2}$$

$$x^T y = 2$$

$$\cos(o_1, o_2) = \frac{2}{\sqrt{3}\sqrt{3}} = \frac{2}{3}$$

Which of the following statements are true?

A. $SMC(o_1, o_2) = \frac{3}{5}$ $J(o_1, o_2) = \frac{1}{2}$, $\cos(o_1, o_2) = \frac{2}{3}$,

~~B. $SMC(o_1, o_2) = \frac{3}{5}$ $J(o_1, o_2) = \frac{3}{4}$, $\cos(o_1, o_2) = \sqrt{\frac{2}{3}}$,~~

~~C. $SMC(o_1, o_2) = \frac{2}{5}$ $J(o_1, o_2) = \frac{1}{3}$, $\cos(o_1, o_2) = \frac{2}{3}$,~~

~~D. $SMC(o_1, o_2) = \frac{2}{5}$ $J(o_1, o_2) = \frac{1}{3}$, $\cos(o_1, o_2) = \sqrt{\frac{2}{3}}$,~~

E. Don't know.

$$SMC(x, y) = \frac{f_{00} + f_{11}}{K}$$

$$J(x, y) = \frac{f_{11}}{K - f_{00}}$$

$$\cos(x, y) = \frac{x^T y}{\|x\|_2 \|y\|_2}$$

$$EJ(x, y) = \frac{x^T y}{\|x\|_2^2 + \|y\|_2^2 - x^T y}$$

$$\|x\|_2 = \sqrt{1^2 + 1^2 + 1^2 + 0^2 + 0^2} = \sqrt{3}$$

$$\|y\|_2 = \sqrt{0^2 + 1^2 + 1^2 + 0^2 + 1^2} = \sqrt{3}$$

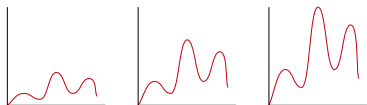
The problem is easily solved by using the inserted formula. We obtain: $\text{SMC}(o_1, o_2) = \frac{3}{5}$ $J(o_1, o_2) = \frac{1}{2}$, $\cos(o_1, o_2) = \frac{2}{3}$ and therefore the A is true. Since the

data is binary, the extended Jaccard and the jaccard coefficient agree.

Invariance

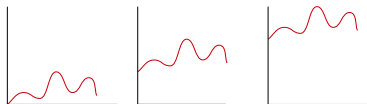
Scale invariance

$$d(\mathbf{x}, \mathbf{y}) = d(\alpha \mathbf{x}, \mathbf{y})$$



Translation invariance

$$d(\mathbf{x}, \mathbf{y}) = d(\alpha + \mathbf{x}, \mathbf{y})$$



General invariances

$$d(\boxed{6}, \boxed{2}) = d(\boxed{6}, \boxed{2})$$

Transformations

Standardization: Ensure a single attribute will not dominate:

$$\tilde{x}_i^{(k)} = \frac{x_i^{(k)} - \hat{\mu}_k}{\hat{\sigma}_k}$$

Example:

- **Number of children** 0-5
- **Age** 0-100 years
- **Annual income** 0-50.000 €

Combining heterogeneous attributes Transform measures and combine

$$s_{\text{Edu.}} = \text{SMC}(x_{\text{Edu.}}, y_{\text{Edu.}})$$

$$s_{\text{Age.}} = a \left(a + d_1(x_{\text{Age.}}, y_{\text{Age.}}) \right)^{-1}, \quad a = 1$$

$$s(x, y) = \frac{1}{2} (s_{\text{Edu.}} + s_{\text{Age.}})$$

Example:

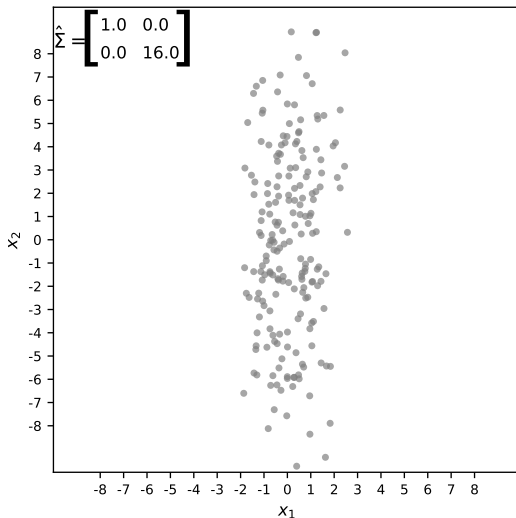
- **Age:** Continuous
- **Education:** Binary
 - Primary (yes/no)
 - Secondary (yes/no)
 - Tertiary (yes/no)

Weighting Attributes have different importance

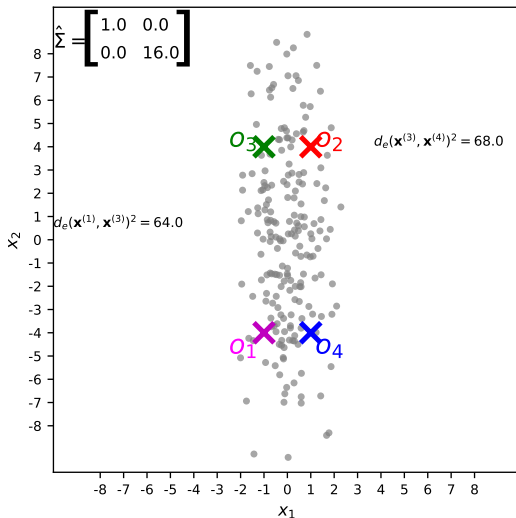
$$s(x, y) = 0.99s_{\text{Edu.}} + 0.01s_{\text{Age.}}$$

Example: What does scale and units mean

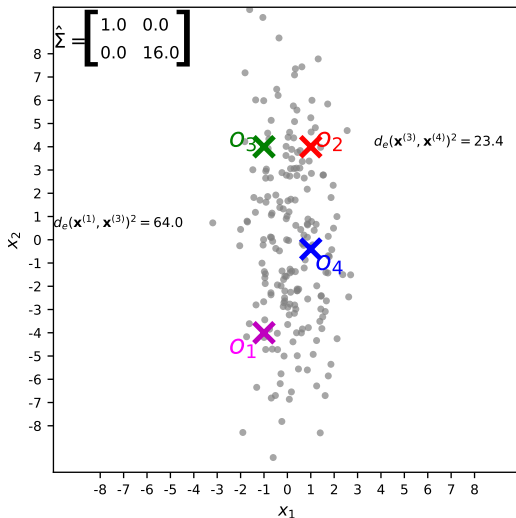
Scale and Standardization



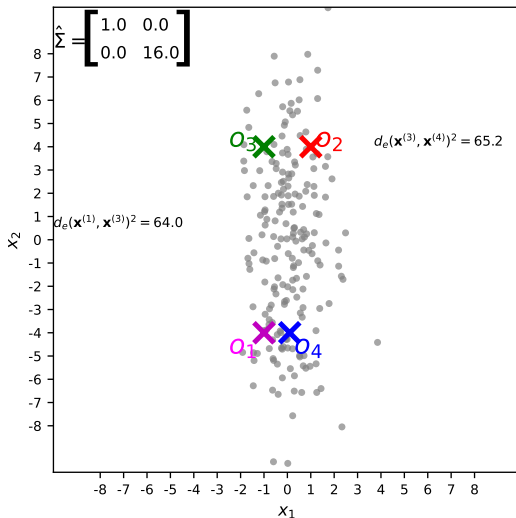
Scale and Standardization



Scale and Standardization

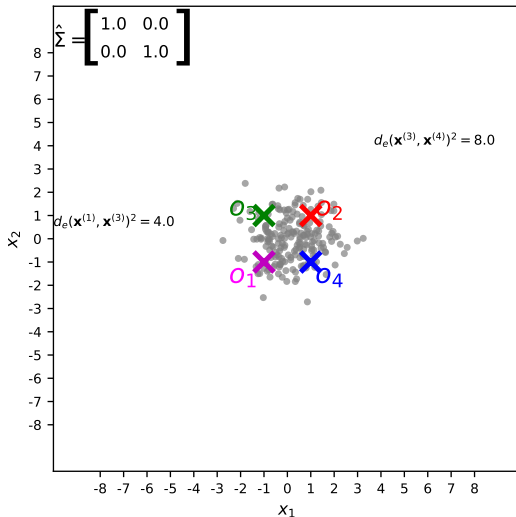


Scale and Standardization

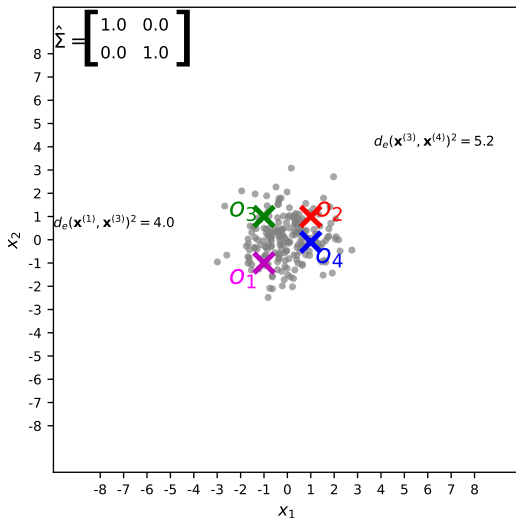


Scale and Standardization

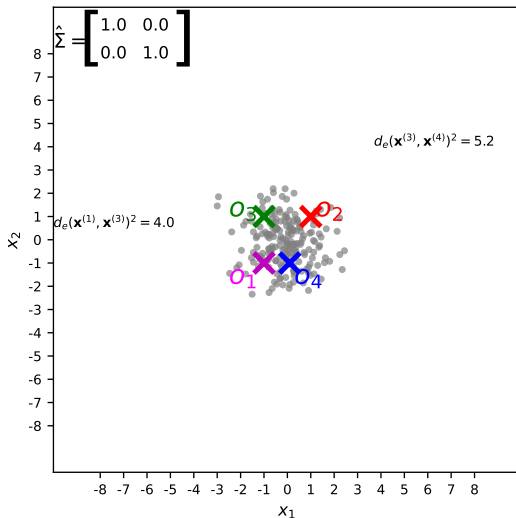
$$\tilde{x}_i^{(k)} = \frac{x_i^{(k)} - \hat{\mu}_k}{\hat{\sigma}_k}$$



Scale and Standardization



Scale and Standardization

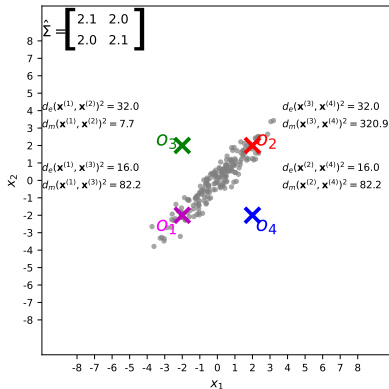


Mahalanobis distance

Define a (squared) distance that takes into account variance and covariance.

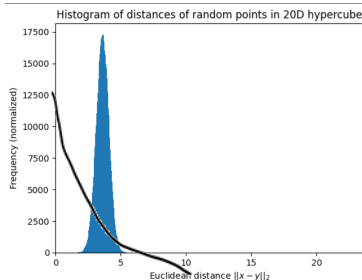
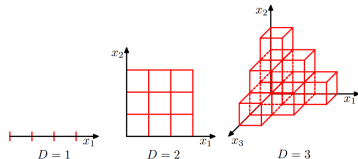
$$d_{euclidian}(\mathbf{x}^{(k)}, \mathbf{x}^{(m)})^2 = (\mathbf{x}^{(k)} - \mathbf{x}^{(m)})^\top \mathbf{I}^{-1} (\mathbf{x}^{(k)} - \mathbf{x}^{(m)})$$

$$d_{mahalanobis}(\mathbf{x}^{(k)}, \mathbf{x}^{(m)})^2 = (\mathbf{x}^{(k)} - \mathbf{x}^{(m)})^\top \mathbf{\Sigma}^{-1} (\mathbf{x}^{(k)} - \mathbf{x}^{(m)})$$



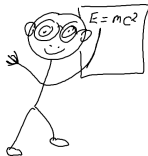
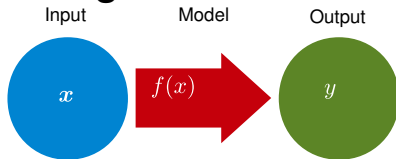
What happens in (very) high dimensions?

- The volume of a unit-cube in high dimensions is 1, but the number of points needed to densely cover the space grows exponentially.
- Imagine a high dimensional box (e.g. 20D), fill it with random points. For any given point, in high enough dimension, the boundaries of the box will be closer than any other point in the box.
- In high dimensions, the ratio of the smallest (Euclidian) distance to the largest distance between points approaches 1. This means that all points become nearly "equidistant" from each other.
- Many more paradoxes to be explored.



K-Nearest Neighbors

Supervised learning



- **Data**

- Inputs and outputs (*this is what we are given*)

$$\{\mathbf{x}_n, y_n\}_{n=1}^N$$

- **Model**

- Function with parameters $\mathbf{w}$ that maps inputs to outputs (*what we are trying to determine*)

$$f(\mathbf{x}; \mathbf{w})$$

- **Cost function**

- Dissimilarity measure between observation and prediction (*how we tell if a model is good or bad*)

$$d(y, f(\mathbf{x}; \mathbf{w}))$$

- **Types** of supervised learning (in this course)

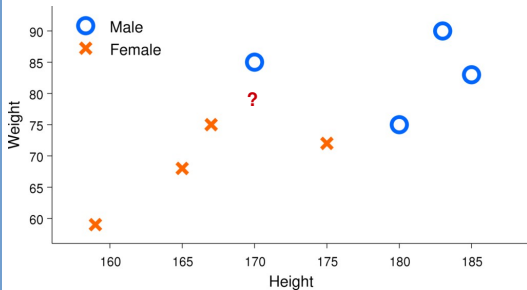
- Regression: Continuous output y
- Classification: Discrete output y

Classification: Definition

- Given a collection of data objects (**training set**)
 - Each object has associated features (aka variables, attributes)
 - Each object belongs to a certain class
- Define a **model** for the class given the other features
- Goal: Assign a class label to a **previous unseen object**

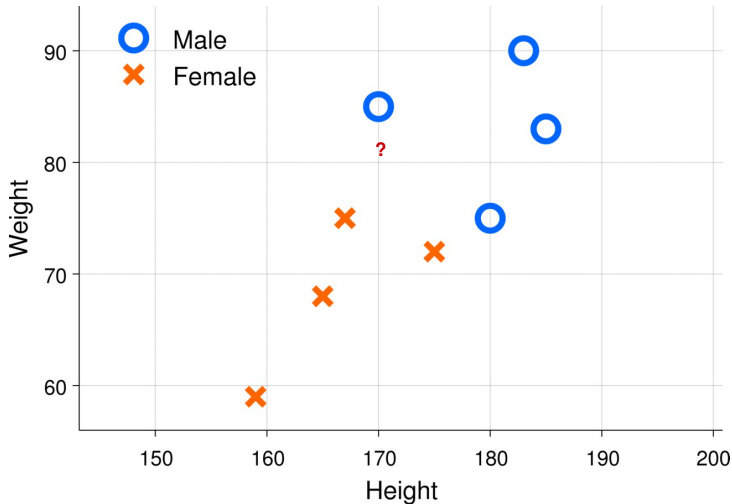
Classify gender based on height and weight

	Height	Weight	Gender
1	183	90	Male
2	180	75	Male
3	170	85	Male
4	185	83	Male
5	159	59	Female
6	167	75	Female
7	165	68	Female
8	175	72	Female
9	171	82	?



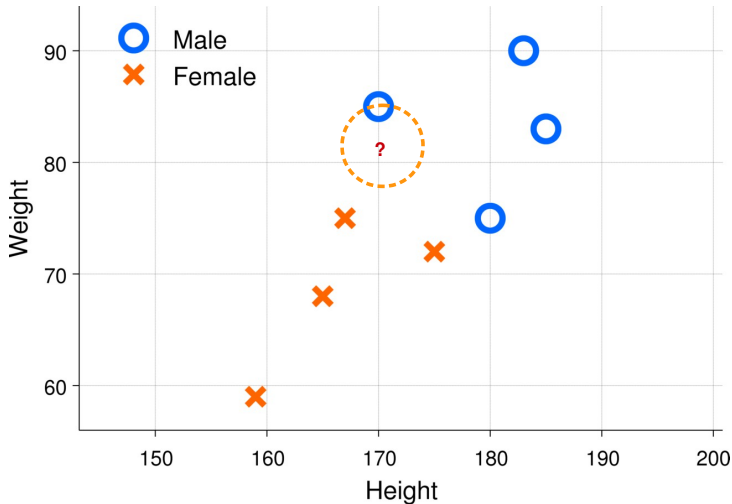
Nearest neighbor classifier

- 1 nearest neighbor



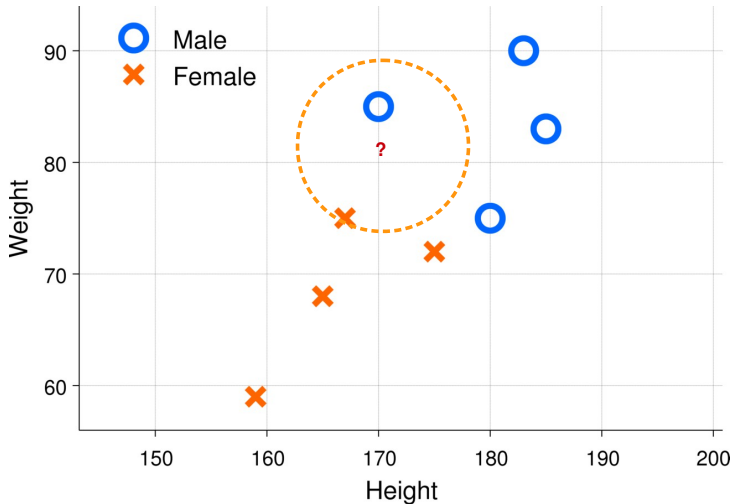
Nearest neighbor classifier

- 1 nearest neighbor



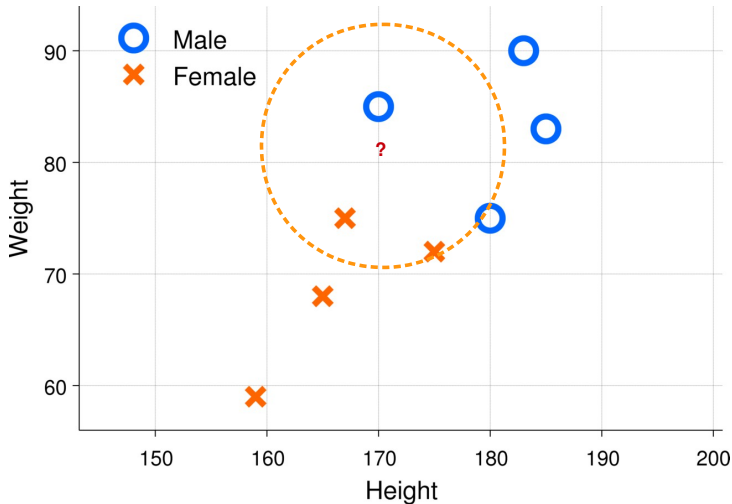
Nearest neighbor classifier

- 2 nearest neighbor



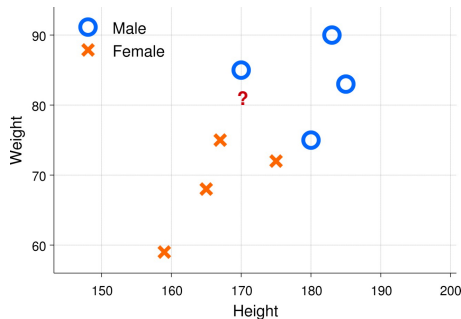
Nearest neighbor classifier

- 3 nearest neighbor

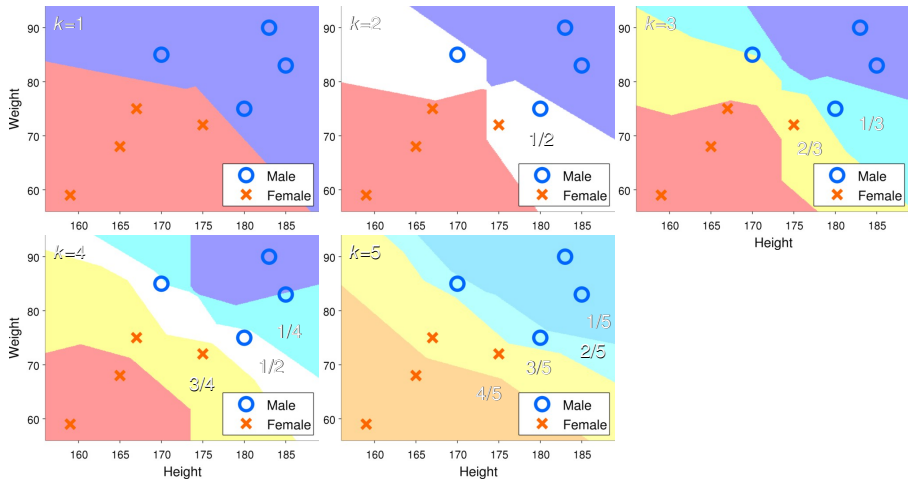


Nearest neighbor classifier

- Choose:
 - The number of neighbors, K
 - A distance measure
- 1 Compute distance to all other data objects
- 2 Find the K nearest data objects
- 3 Classify according to majority of neighbors



Nearest neighbor decision surface



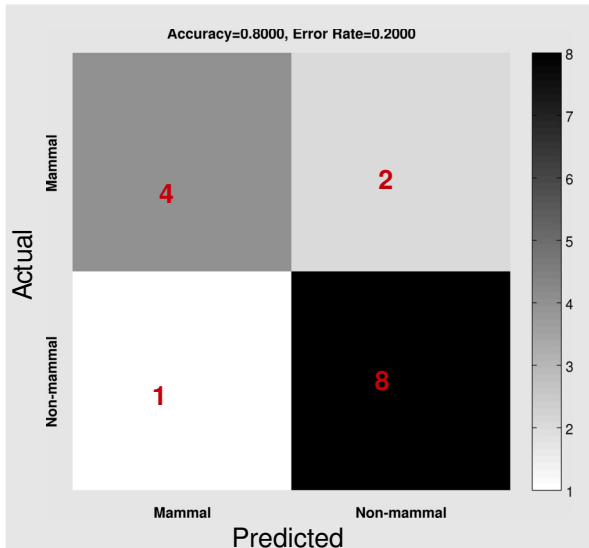
Evaluating a classifier

- Visualization of actual versus predicted class labels
- Accuracy
Number of correctly predicted observations divided by the total number of observations

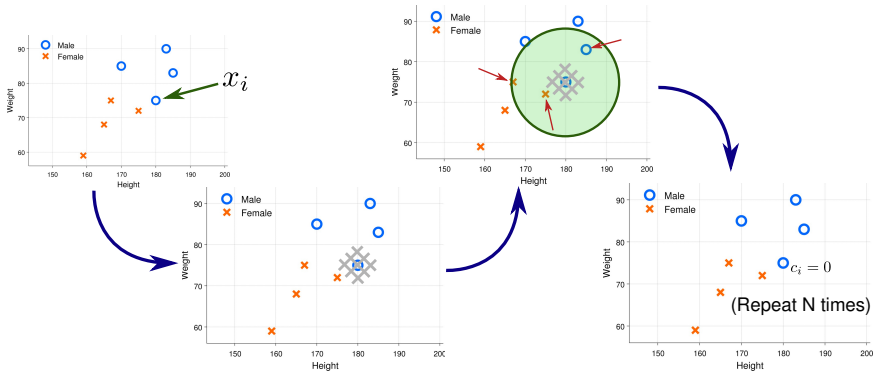
$$\frac{4 + 8}{4 + 2 + 1 + 8} = 0.8 \text{ (80\%)}$$

- Error rate
Number of incorrectly predicted observations divided by the total number of observations

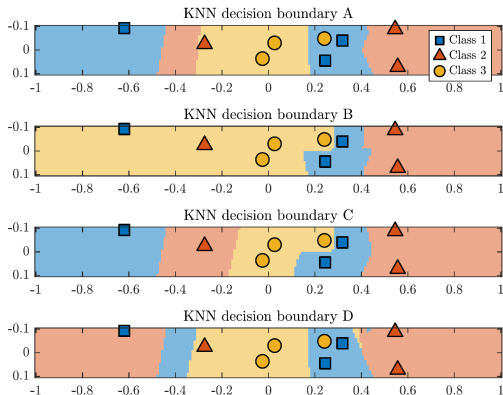
$$\frac{2 + 1}{4 + 2 + 1 + 8} = 0.2 \text{ (20\%)}$$



Evaluating a KNN classifier on "unseen" data



Quiz 3: KNN boundaries



Consider a two-dimensional dataset comprised of $N = 9$ observations shown in Figure 12. The dataset consists of three classes indicated by the blue squares (class 1), red triangles (class 2) and yellow circles (class 3). In the figure, the decision boundaries for four K -nearest neighbor classifiers (KNN) are shown. Which one of the plots correspond to the $K = 3$ nearest-neighbour classifier assuming ties are broken by assigning the point to the *nearest* neighbour's class?

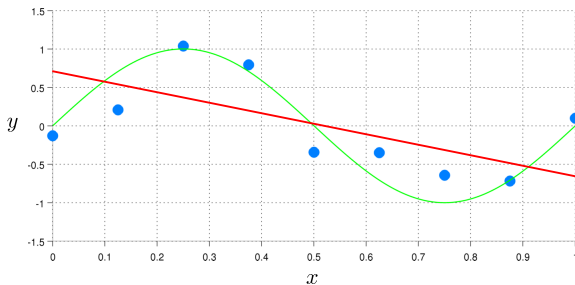
- A. KNN decision boundary A
- B. KNN decision boundary B
- C. KNN decision boundary C
- D. KNN decision boundary D
- E. Don't know.

Quiz 03: KNN boundaries - solution

- KNN decision boundary A, is the correct solution with $K = 3$.
- KNN decision boundary D, here $K = 3$ but for the tie-break we use the label of the second nearest point.
- KNN decision boundary B, here $K = 4$.
- KNN decision boundary C, here $K = 1$.

Regression

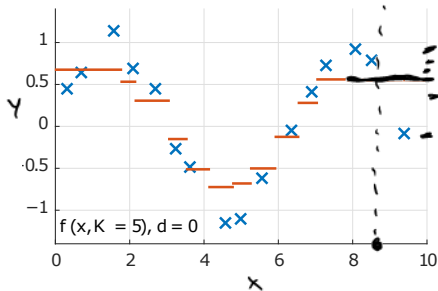
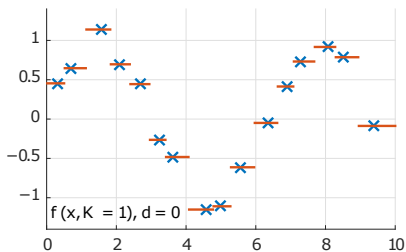
- **Definition:** Learning a function, $f(x)$, that maps a data object to a continuous-valued output
- **Why Regression?**
 - Descriptive modeling
 - Explain/understand the relation between attributes and continuous-valued output
 - Predictive modeling
 - Predict the output value of a new data object
- **Methods:** KNN regression. *Later:* Linear regression, regression trees. regularised linear regression/ridge regression, neural networks, ensembles.



KNN Regression

- Given a training set $\mathbf{X}, \mathbf{y}$
- For a test observation x predict the average y -value in the neighbourhood

$$\hat{y} = f(\mathbf{x}, K) = \frac{1}{K} \sum_{i \in N_{\mathbf{x}}(\mathbf{x}, K)} y_i$$



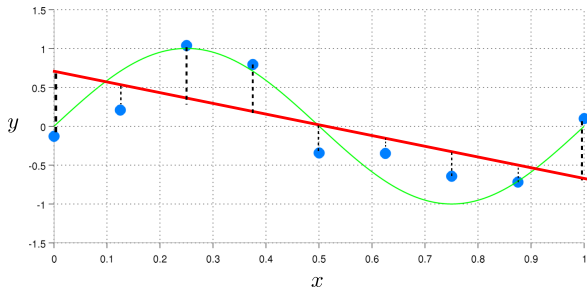
Evaluating a regression model

Compute average loss per observation:

$$E = \frac{1}{N} \sum_{i=1}^N L(y_i, f(\mathbf{x}_i))$$

Where we either use L_1 or L_2 (Euclidean) loss

$$L_1(y_i, \hat{y}_i) = |y_i - \hat{y}_i|, \quad L_2(y_i, \hat{y}_i) = (y_i - \hat{y}_i)^2 \quad (\text{Compare these to } p\text{-norms})$$



KNN methods - open challenges (for now)

- Model selection:
 - How do we systematically select the distance measure?
 - How do we systematically select K?
- Evaluation
 - How do we properly evaluate on "unseen" data?

Summary

Resources

<https://www.3blue1brown.com> An great, animated recap of linear algebra

(<https://www.3blue1brown.com/essence-of-linear-algebra-page/>)

<http://www2.imm.dtu.dk> Our demo of the multivariate normal distribution
which illustrates the effect of the covariance matrix

(<http://www2.imm.dtu.dk/courses/02450/DemoNormal.html>)